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ACT-R Φ : A Cognitive Architecture with Physiology and Affect

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Abstract

Current computational cognitive architectures lack a comprehensive representation of human physiology. Connecting a more realistic model of physiology to a cognitive architecture allows one to develop more realistic computational process models that include an account for the effect of physiology on cognition. It is also useful to include some theory of affect and emotion to act as a functional layer between some representations of physiology and cognition. We chose to extend the ACT-R (adaptive control of thought – rational) cognitive architecture with physiology and affect to make the ACT-R Φ hybrid architecture. We connected the HumMod physiological simulation system with the ACT-R architecture; ACT-R Φ has a module that communicates between ACT-R and HumMod. A new affect system was also developed using an existing theoretical framework to serve as a functional layer between some of the physiological and cognitive representations. With this new architecture one can model a wide range of human behavior (e.g., the effects of homeostasis on cognition) in a more tractable fashion.

Keywords: Cognitive Architecture, ACT-R, Physiology, HumMod, Affect, Cognitive Model

Introduction

Current computational cognitive architectures lack a comprehensive representation of human physiology. Having a more realistic representation of human physiology within a cognitive architecture is important as it allows one to model both the physiological and psychological aspects of an experiment both of which may have significant effects on one another. As one develops connections between the physiological and cognitive levels in a computational cognitive architecture it is useful to use a theory of affect or emotion to act as an intermediate layer between the two since the effect of cognition on physiology (or vice versa) is often characterized in terms of emotion or affect (e.g., Berridge et al., 2009; Craig, 2008; Damasio & Carvalho, 2013; LeDoux & Phelps, 2008; Panksepp & Biven, 2012).

With an architecture-based framework for the representation of affect and emotion, one can represent human behavior more realistically. While several cognitive architectures can represent many aspects of human cognition and behavior, many of the details available from literature in affect and emotion remain

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difficult to represent (see Bach, 2009; Marsella et al., 2010; Ritter et al., 2012; Silverman et al., 2012 for examples of other implementations of representations related to emotion). Adding affective or emotional behavior by extending an underlying architecture also matches more closely with recent work on finding associations between ACT-R and brain imaging (e.g., Anderson et al., 2008) given the amount of affect and emotion literature that ties neural correlates to certain behavior. The architectural approach also has the potential to contribute to understanding as one can simulate how the integration of these functional processes can continually affect human behavior.

ACT-R Φ is a hybrid computational architecture that adds a model of a physiological substrate and a representation of affect to the ACT-R (Anderson, 2007) cognitive architecture. The HumMod simulation system (Hester et al., 2011) is used for the physiological substrate while primary-process affect theory (Panksepp & Biven, 2012) is used to develop a functional layer between some of the cognitive and physiological representations. This new computational system gives one the opportunity to model the effect that human physiology and cognition have on one another with higher tractability.

HumMod

HumMod (Hester et al., 2011) is a simulation system that provides a top-down model of human physiology (e.g., organs, hormones, etc.). The model of physiology is a derivative of the original Guyton cardiovascular model (Guyton et al., 1972). This derivative is much more expanded than the Guyton model (e.g., it now includes representations hormones and catecholamines) and has the advantage of being written in XML. Though the underlying XML model is available to change, the easiest entry points to alter the model are the model's specified parameters (e.g., changing the *IVDrip.Switch* parameter to simulate the insertion of an intravenous line). HumMod also allows one to run the physiological model over multiple time-scales and to change parameters while running the model within the simulation.

The physiology model contains over 1,500 linear and non-linear equations and over 6,500 state variables (Table 1). As few variables have a one-to-one connection, the use of parameters to change the values of these variables allows another computational system (e.g., ACT-R) to alter the model with few unintended consequences.

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Table 1: Some of HumMod's physiology systems

HumMod System	#Vars	Example Vars
Body Fluids	214	Osmolarity (Cell Wall)
Circulation	426	SA Node Rate
Electrolytes	140	Blood Lactate Pool
Hormones	534	Adrenocorticotrophic Secretion
Metabolism	321	Glucose Pool Mass
Nervous System	187	Norepinephrine Pool Mass
Organs	2,349	GI Alpha Receptors
Respiration	326	Breathing Tidal Volume
Other Systems (Lifestyle, Heat, etc.)	2,026	Skin Temperature

Primary-Process Affect Theory

Panksepp and Biven (2012) posit affective systems in the brain based on coherent unconditioned behavior in response to stimulation of certain neural circuitry. These neural systems operate on the “primary-process” level, i.e. below the level of learning mechanisms. The most basic of these primary circuits is the SEEKING system that accounts for appetitive motivation and very closely aligns with existing incentive salience (or wanting) theory (Berridge et al., 2009).

The similarities between the neural circuitry and corresponding functionality of the SEEKING and wanting is very useful as the two tend to be describe in terms of different levels. SEEKING literature (and related experiments) tends to focus the underlying motivational and affective component that can drive behavior, while wanting/incentive salience literature tends to also include learning systems (e.g., reinforcement learning). Viewing the two as complementary allows one to use the two theories to inform the development of an affective system within a computational cognitive architecture.

ACT-R

ACT-R (Anderson, 2007) is a modular cognitive architecture that allows for symbolic and subsymbolic processing. Modules in ACT-R function in parallel, but operate in serial when communicating with the production system, which includes procedural memory. ACT-R’s modularity combined with the open-source nature of the software makes the architecture very easy extend as one can add modules to the architecture and have the function automatically loaded within the ACT-R software system.

The functionality of ACT-R modules has also been associated with brain activity data found during functional brain imaging experiments (e.g., Anderson, 2007; Anderson et al., 2008). This cross between

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theoretical (ACT-R) and empirical (fMRI) cognitive science allows one to more easily connect existing theories and experimental data in psychology and neuroscience that use brain-imaging data.

Extending ACT-R with Physiology and Affect

We have extended a previous version of the ACT-R Φ hybrid architecture (Dancy et al., In Press) with representations of affect, that is, those given by Panksepp and Biven (2012). This new extension gives ACT-R Φ a functional layer between the physiological and cognitive systems that is based on existing neuroscience and psychological evidence; it also allows one to simulate some effects of homeostasis on cognition (e.g., the effect of being thirsty on one's choices and decision-making).

Using the primary-affect theoretical framework to construct a functional affect level between the cognitive and physiological levels is attractive as primary-process affect theory focuses particularly on experimental data from neuroscience and attempts to separate different levels of neural and behavioral function (e.g., primary-process, secondary-process, tertiary-process; Panksepp & Biven, 2012). When one considers existing work in the closely related incentive-salience, or wanting, framework (e.g., Berridge et al., 2009; Smith et al., 2011; Zhang et al., 2009) that tends to take a secondary-process (that includes learning processes) perspective, one is left with neural systems related to different psychological functionality that can be associated with ACT-R brain-imaging work (e.g., Anderson et al., 2008; Möbus et al., 2011; Stocco & Anderson, 2008); Figure 1 displays some key brain areas from the three theoretical frameworks.

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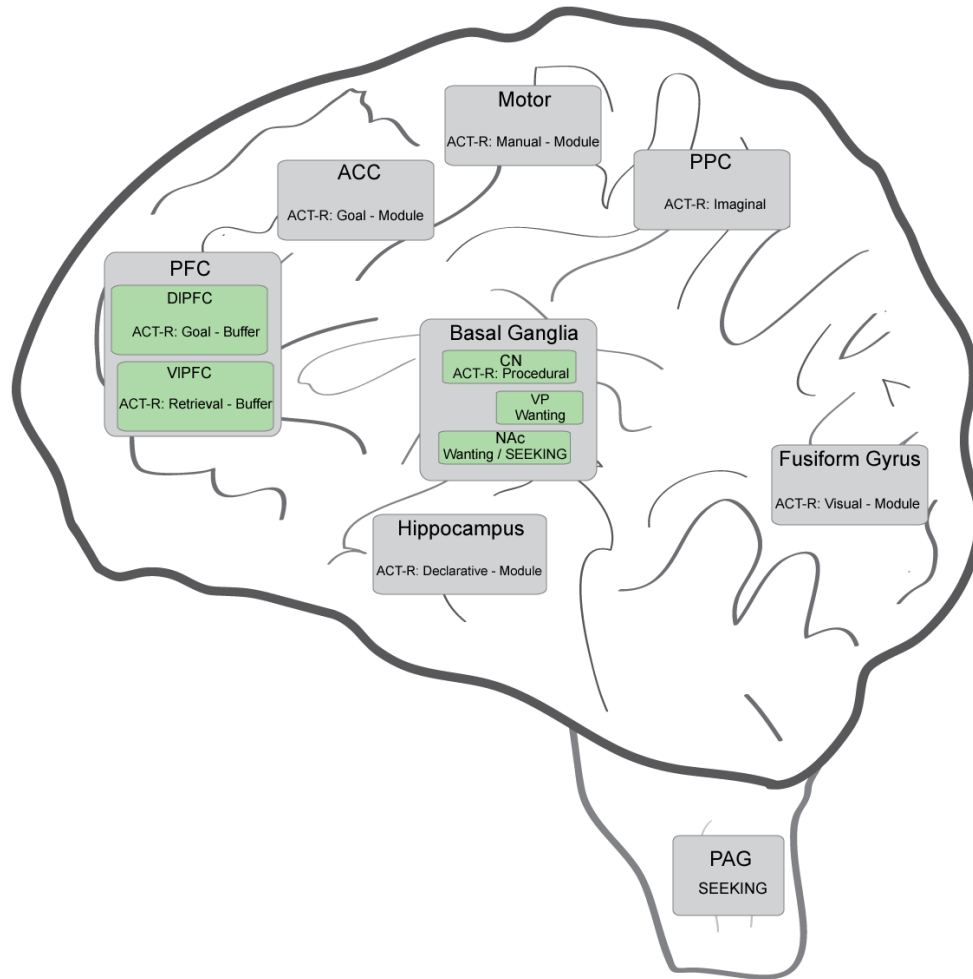


Figure 1: Some key brain areas from the SEEKING, incentive salience, and ACT-R theories

ACT-R Φ (Figure 2) uses modules to provide representations of physiology and affect. Extending ACT-R using its existing module system had two particular advantages: (1) we were able to develop a hybrid architecture that would have functionality familiar to those who've used the traditional ACT-R cognitive architecture (e.g., using the *sgp* function to change module parameter values); (2) models that were developed to run within the ACT-R architecture should still be able to run within the new architecture. We extended ACT-R with the *physio*, *SEEKING*, and *Affective-Associations* modules. The *SEEKING* and *Affective-Associations* modules are contained within the affect system in Figure 2.

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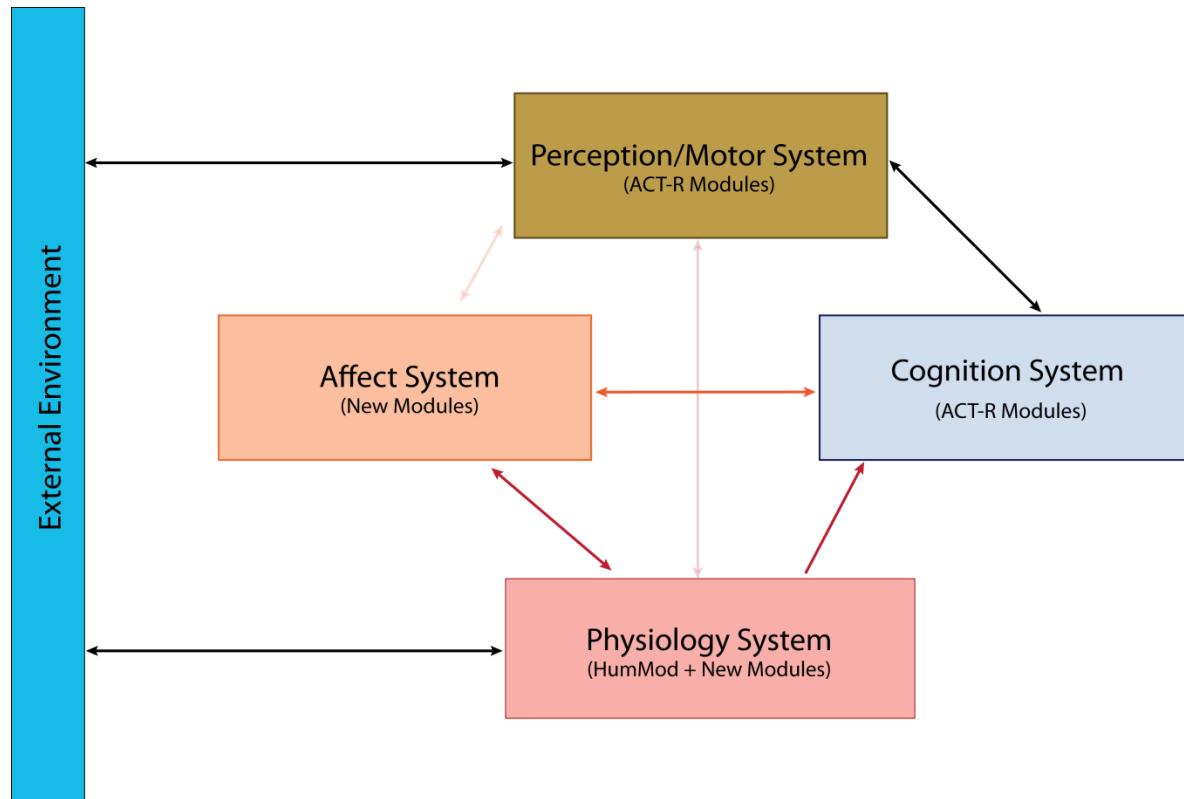


Figure 2: A diagram of the ACT-R Φ subsystems

The physio module acts as a communication layer between the original ACT-R architecture and the HumMod system. As the HumMod simulation is run physiological values are stored in (simulated) real-time within the physio module. There are functions within the module that allow a modeler to change HumMod variables and get the values of those variables or the course of any model run. The *set-phys-var* function and *get-phys-var* functions can be passed a list of name/value sets specify which variable will be changed or requested. The module can also be set (via a parameter) so the HumMod system is ignored and the modeler is essentially interacting with the original ACT-R cognitive architecture.

SEEKING (J. Wright & Panksepp, 2012) describes appetitive affect that underlies basic approach behavior. This neural system is very similar to wanting motivation theorized by (Berridge et al., 2009) that is often described from the perspective of learning, i.e. incentive salience (Berridge et al., 2009; Smith et al., 2011; Zhang et al., 2009). Especially useful is the more recent work in separating hedonic liking, and motivation-based wanting into two different constructs that are described separately in reinforcement equations (Zhang et al., 2009). Following this, we add an offset to the ACT-R utility function that represents the bottom-up dynamic modulation of the SEEKING (or wanting) system (equations 1, 2, and 3).

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$$(1) \quad U_A = U + \log(k)$$

$$(2) \quad k = sVal * e^{reward_{max}} \text{ s.t. } sVal \in [0,1]$$

$$(3) \quad sVal_{Thirst} = \frac{reward_{max} * (osmo - osmo_{base})}{osmo_{base} - osmo_{min}} + noise$$

Variable k is some modulating factor (e.g., homeostatic change) that causes dynamic change in utility/value and $sVal$ is some value that depends on a specific (behavioral) function. Equation 3 is the $sVal$ function for thirst-based homeostatic affect that uses the model's current physiological thirst, which is determined by osmolarity, to find a thirst value. There are currently four $sVal$ -based functions (hunger, thirst, skin-temperature, goal) and the output from all functions is normalized to some value between 0 and 1 (inclusive) to find the function with the highest corresponding value. The highest $sVal$ is applied to the k function (equation 2) and this value is then added to any existing utilities that have been assigned to that particular function using the *affective-associations* module. Given ACT-R's reward equation, integrating this modulation into the learned reward of some set of productions using equation 4 becomes less clear. The current default reward-learning mechanism (equation 5) is different from the version often seen in reinforcement learning literature. This is because the more neurally-inspired version (equation 6) can be problematic for certain models that use it (e.g., Fu & Anderson, 2006).

$$(4) \quad U_i(n) = U_i(n-1) + \alpha(R_i(n) - U_i(n-1))$$

$$(5) \quad R_i(n) = r_j - (t_j - t_i)$$

$$(6) \quad R_i(n) = r_i + \left(\frac{1}{1+kt}\right) * U_{i+1}(n-1) - U_i(n-1)$$

In equation 4, $U_i(n)$ is the utility of production i and α is some learning rate that is an available parameter in ACT-R. Reward in equation 5 is determined by the reward triggered minus the difference between the time of the current production and the time of the production fired directly after the reward was triggered. In equation 6 (the problematic neurally-inspired reward equation), the utility of the production previously fired, $U_{i+1}(n-1)$ is used to determine the reward instead of a simple time difference. One would likely want to add a new modulating variable (e.g., k in equation 1) as an offset to the reward to represent a learning of the motivating factor, however one still may run into the problems found by Fu and Anderson (2006) when attempting to use equation 6.

The affect-associations module houses functionality that allows the two different ACT-R memory systems (procedural and declarative memory) to be associated with different affects. We chose to develop a separate module for this functionality due to existing evidence that affective association happens in distinct circuits (e.g., McGaugh, 2004; Seymour & Dolan, 2008). This module allows a modeler to hard-

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code affect labels into certain productions using the *spp-a* function, an extension of the ACT-R *spp* function.

A model developed to use the ACT-RΦ Systems

The new hybrid architecture has been used to develop a model with representations of changes in physiology and corresponding homeostatic affect. We have used the architecture to model of behavior found in an experiment where participants were artificially induced to be thirsty and then asked to complete a task related to their current thirsty state.

We developed a model to complete a version of the experiment carried out by N. D. Wright et al. (2012). In the experiment, participants were artificially induced to be thirsty (via Intravenous administration of hypertonic saline) and then asked to complete a version of the ultimatum game¹ where the reward was primary, i.e. water, instead of monetary. During the task, participants were given a proposal of water reward distribution by some *proposer* (87.5% of water reward to the proposer and 12.5% to the responder); the participants then could either accept or reject the distribution. Participants were only asked to make this choice once during the experiment. Despite an increased osmolarity and self-rated subjective thirst, half of the users still chose to reject an offer for water; however, those who accepted the offer did have a higher rated subjective thirst.

Our model (Figure 3) used the physiological substrate to simulate the induction of thirst by changing the *IVDrip.ClinicalSaline* and *IVDrip.H2OSetting* physiological parameters (set to 860 and 25, respectively) to simulate the effect of hypertonic saline in the HumMod system. Turning on the hypertonic saline IV resulted in a change in osmolarity, which is directly connected to the thirst SEEKING function mentioned in the previous section. This thirst function value then is added to the utility of a production using the equations presented in the previous section. The model produced subjective thirst (sVal) and osmolarity values comparable to those found in the original experiment (Table 2). However, the model did fail to have an acceptance rate similar to the rate found in the original experiment (79% for the model versus 50% for the original experiment). The likely reason for this is the fairly simple and rigid fairness reasoning used by the model (*if offer < x% then reject*). The original experiment also suffered from a lower than desired amount of participants likely because the experimental condition of the task, i.e. intravenously administering high salt content saline, was physiologically demanding; this lower number

¹ The ultimatum game (Güth et al., 1982) is a task where an existing reward (usually monetary) must be split between a *proposer* and a *responder*. The proposer offers a certain distribution of the reward (e.g., 50/50) to a responder who must decide whether or not to accept the proposition. If the responder rejects, then neither participant receives some of the reward, so the proposer often tries to keep the proposed split in their favor while also not making the split so one-sided that the responder rejects the offer for being too unfair.

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of participants could have contributed to some of the discrepancy between our model output and experimental data.

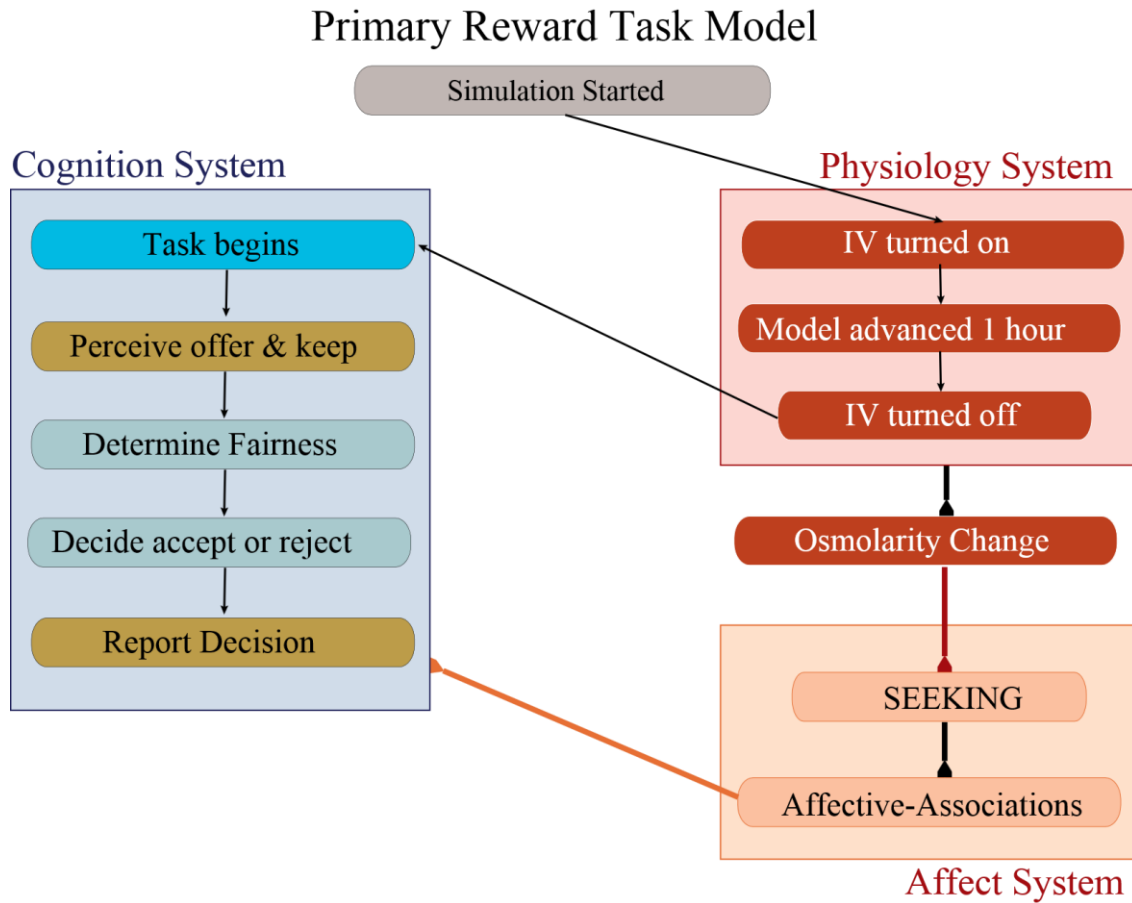


Figure 3: Primary-Ultimatum Task Model. Lines that are thicker represent continuous communication between the two objects.

Table 2: Table of primary model results

Decision	Osmolarity (sd)	Subj. Thirst (sd)
Accept _{experiment}	Not Reported	8.90(1.7)
Accept _{model, n=790}	306.37(0.2)	7.94(1.2)
Reject _{experiment}	Not Reported	5.60(1.6)
Reject _{model, n=210}	305.86(3.74)	4.82(1.2)
Both _{experiment}	310(5.0)	7.30(1.6)
Both _{model, n=1000}	306.27(1.7)	7.29(1.7)

We have briefly discussed a way one can use the ACT-R Φ architecture to model how physiology can affect human behavior that would have been difficult to model in a realistic and tractable manner in other existing architectures. Being able to develop models like these are also important because they allow a

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modeler to develop predictions for physiological effects that would be very difficult to ethically test experimentally. Though the models do represent an important step in developing more unified theories of cognition, i.e. with physiology and affect/emotion, there still remains room for expansion and exploration. We briefly discuss a few potential directions for this work including an expansion of the affect system and more direct connections between physiology and cognition in the architecture.

Future Directions and Conclusion

The SEEKING neural system is the most fundamental of the affective systems posited by Panksepp and Biven (2012). However, given the body of work, the Affect system in ACT-R Φ should be expanded to include some of the other neural systems discussed by Panksepp and Biven. The most obvious expansion would be the FEAR affective system as the behavior caused by this system continues to be widely studied, i.e. studies involving what is often categorized as fearful behavior. This system would need to be tied into the ACT-R production and declarative memory systems in some of the same ways as the SEEKING system.

It is also likely that there should be more direct connections between the physiology system and the cognitive modules in ACT-R Φ . These direct connections would be more representative of some of the global cognitive changes one may see with increased activity in brain areas; an example would be activity in the locus coeruleus and nucleus tractus solitarius that would result in an increase in neural norepinephrine and cause a cascade of changes in different areas of the brain (e.g., Joëls & Baram, 2009).

We have presented a novel extension of the ACT-R architecture that adds representations of physiology and affect, as well as a model that have been developed to use these new representations. As experimental methods that gather both physiological/biological and psychological data become less invasive, it will be important to expand existing computational architectures to allow quantitative process-models with both physiological and cognitive representations. These models will not only be able to quantify theories, but also provide predictions for how physiology may affect cognition and behavior when experimental analysis would be difficult due to ethical considerations. ACT-R Φ is a novel hybrid architecture that can potentially allow these more unified process models to be developed.

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